Lecture 40 — SVD, Lab 6, Quiz 10, and capstone launch

Fri Dec 4

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: selected slides from the Strang/MIT SVD slides (not all 17 pages), Lab 6, capstone launch, Lachniet Ch. 5 (SVD) and Ch. 7 (image compression). Related objectives: Lectures. Schedule.

Packed Friday — keep the roles straight:

Piece	What it is today
Quiz 10	Week 13 **eigenvalues** / Lab 5 — **not** least squares, **not** SVD
Lab 6	Least-squares software (impossible / overdetermined equations)
This short lecture	SVD conceptual + rank-k idea
Capstone	Launch pointer only — studio Mon Dec 7

Bring paper and a laptop. Quiz first (or as scheduled), then a short SVD launch, then Lab 6; capstone details continue next week.

Learning objectives

By the end of class, students should be able to:

1. Interpret $A = U\Sigma V^\mathsf{T}$ conceptually.
2. Construct a rank-k approximation computationally.
3. Compare singular-value decay for an image/frame and a frame-difference matrix.
4. Encode three frames either independently or as F_1, D_2, D_3.
5. Compare reconstruction error and storage for the two approaches.
6. Formulate a capstone question involving accuracy, storage, runtime, and resource use.

(Objectives 3–6 are launched today and carried in the capstone packet; Lab 6 is least squares, not SVD coding.)

Warm-up (after quiz)

A grayscale image (or one video frame) is an $m \times n$ matrix of pixel intensities.

(a) Roughly how many numbers store the full matrix?

(b) If most "information" lived in a few patterns, why might a factored low-rank form save storage?

Fill in

SVD (conceptual). Every real $m \times n$ matrix factors as

$$A = U\Sigma V^{\mathsf{T}},$$

where the columns of U and V are orthonormal (left / right **singular vectors**) and Σ is rectangular "diagonal" with nonnegative **singular values**

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$$

on its main diagonal. Think: A = rotate/reflect $\rightarrow$ stretch by σ_i $\rightarrow$ rotate/reflect.

Rank-k approximation. Keep the largest k singular values (and matching singular-vector columns); set the rest to zero. The truncated product

$$A_k = U_k \Sigma_k V_k^{\mathsf{T}}$$

is the best rank-k approximation in the usual 2-norm / Frobenius sense (details in the slides / capstone).

Software sketch (capstone / later — not Lab 6 today):

Task	Octave	Python
SVD	`[U,S,V] = svd(A)`	`U, S, Vt = np.linalg.svd(A, full_matrices=False)`
Rank-k rebuild	use first k columns / σ's	`U[:,:k] @ np.diag(S[:k]) @ Vt[:k,:]`

Capstone launch (one question). For a short clip of frames: when is it better to compress each frame alone versus encode F_1 and differences $D_2 = F_2 - F_1$, $D_3 = F_3 - F_2$? Trade off **accuracy**, **storage**, **runtime**, and **resource use** — including the cost of computing the SVD itself. Full prompts live in the capstone packet; studio is Mon Dec 7.

Worked in class (short)

W1. Read Σ. Suppose a 4×3 matrix has singular values $\sigma = (10, 2, 0.1)$. What is rank(A) (numerically "full" vs tiny)? For a rank-1 approximation, which σ do you keep?

W2. Storage cartoon. Full $m \times n$ needs mn entries. A rank-k factorization stores about $k(m + n + 1)$ numbers (columns of U_k, V_k, and k singular values). When is $k(m + n + 1) \ll mn$?

W3. Frames vs differences (launch). Three frames F_1, F_2, F_3. Plan A: compress each. Plan B: compress F_1, $D_2 = F_2 - F_1$, $D_3 = F_3 - F_2$. In one sentence each: when might Plan B win on storage? What must you still check (error / runtime)?

Your turn

Y1. In $A = U\Sigma V^{\mathsf{T}}$, what do the singular values measure conceptually (one phrase)?

Y2. True or false: computing an SVD is always "free," so low-rank multiplication is always faster overall.

Y3. Same-day sorting: which of Quiz 10, Lab 6, and the capstone launch assesses Week 13 eigenvalues?

Y4. Write one capstone question you care about (accuracy, storage, runtime, or resource use) for frames vs frame-differences. One sentence is enough — expand in studio.

Lab 6 and capstone, same day

After the short SVD launch: Lab 6 — Impossible Equations (least-squares software from Lectures 38–39). Preview the one-pager; write on the hard copy in class.

Capstone: skim the capstone packet launch objectives; full studio Mon Dec 7. Do **not** expect to finish image/SVD experiments today.

Reminder: Quiz 10 covers Week 13 (eigenvectors / characteristic equation / Lab 5), **not** least squares and **not** today's SVD notes.

Answers (Your turn)

Y1. Stretch factors / importance of each singular-vector mode (how strongly A amplifies that direction).

Y2. **False.** The factorization itself can be expensive; low-rank reuse only pays off when approximation quality is good **and** you reuse the factors enough (see capstone notes).

Y3. **Quiz 10** only.

Y4. Any clear tradeoff question is fine — e.g. "For my three frames, does difference encoding cut storage at the same relative error, after counting SVD time?"